

Competitive Exam Practice 3: IEEE 754 Floating-Point Representation and Arithmetic

GATE (Computer Science) Pattern

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Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type.

Q1 [*GATE CS 2014 Set 2 Q45, verified — GATEOverflow V3 p.331*] [**MCQ, 1 mark**] A float variable X is assigned the decimal value -14.25 , represented in IEEE-754 single-precision (1 sign + 8 exponent + 23 mantissa). The representation of X in hexadecimal notation is:

- (A) C1640000H
- (B) 416C0000H
- (C) 41640000H
- (D) C16C0000H

Q2 [*GATE CS 2017 Set 2 Q12, verified — GATEOverflow V3 p.331*] [**MCQ, 1 mark**] Given the following 32-bit single-precision IEEE-754 number: 0011 1110 0110 1101 0000 0000 0000 0000. The decimal value closest to this floating-point number is:

- (A) 1.45×10^1
- (B) 1.45×10^{-1}
- (C) 2.27×10^1
- (D) 2.27×10^{-1}

Q3 [*GATE CS 2023 Q35, verified — GATEOverflow V3 p.332*] [**MCQ, 2 marks**] Consider the IEEE-754 single-precision numbers $P = 0xC1800000$ and $Q = 0x3F5C2EF4$. The product $P \times Q$, represented in IEEE-754 single precision, is:

- (A) 0x404C2EF4
- (B) 0x405C2EF4
- (C) 0xC15C2EF4
- (D) 0xC14C2EF4

Q4 [*GATE CS 2008 Q4, verified — GATEOverflow V3 p.330*] [**MCQ, 1 mark**] In the IEEE floating-point representation, the hexadecimal value 0x00000000 corresponds to:

- (A) The normalized value 2^{-127}

- (B) The normalized value 2^{-126}
- (C) The normalized value $+0$
- (D) The special value $+0$

Q5 [*Original, GATE-pattern*] [MCQ, 1 mark] In IEEE-754 single precision, the exponent is stored in biased form ($E_{\text{stored}} = E_{\text{true}} + 127$). The reason for the bias, rather than two's complement, is that:

- (A) two's complement cannot represent negative exponents
- (B) it makes the exponent field all zeros correspond to the smallest exponent and all ones to the largest, so unsigned comparison orders floats correctly
- (C) it doubles the mantissa precision
- (D) it eliminates the need for normalization

Q6 [*Original, GATE-pattern*] [MCQ, 2 marks] A loop computes $x = 0.1$ ten times by repeated addition in single-precision floating point, then tests $x = 1.0$. Which statement is **TRUE**?

- (A) The test always passes because $10 \times 0.1 = 1.0$ exactly
- (B) The test always fails because the product of ten adds is exactly $0.999\dots$
- (C) The test may fail because 0.1 is not exactly representable and rounding error accumulates across the additions
- (D) The test fails only if overflow occurs